ST MARY'S DSG, KLOOF

GRADE 10 MATHEMATICS

Prop Int Theorem, Inequalities, Number Systems and Straight lines

May 2023

TIME: 80 MINUTES **TOTAL: 66 MARKS**

NAME: _____ **MATHS TEACHER:_____**

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READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of 9 pages and 7 questions.
2. Answer all the questions on the question paper.
3. All the necessary working details must be clearly shown.
4. Approved non-programmable calculators may be used except where otherwise stated.
5. It is in your own interest to write legibly and to present your work neatly.

QUESTION 1

Given the numbers :

27 −3 $\frac{2}{25}$ 3

$$3^3 \; ; \; -\sqrt[3]{27} \; ; \; 30 \; ; \; 3\pi \; ; \; \sqrt{\frac{4}{25}} \; ; \; \sqrt{3} \; ; \; \sqrt{-4} \; ; \; -2,\dot{1}\dot{4} \; ; \; (\sqrt{3})^2$$

a) Which of these numbers are :

1. An integer with factors 2 and 5.
   30
2. Rational numbers that are not integers.
   $\sqrt{\frac{4}{25}}$ $-2,\dot{1}\dot{4}$
3. An integer, but not a counting number.
   $-\sqrt[3]{27}$
4. Non real number/s
   $\sqrt{-4}$
5. All the irrational numbers. 3π $\sqrt{3}$

(5)

b) Given : $A = \sqrt{4x - 3}$

Give ONE value of x that satisfies the condition for A.

1. $A = 0$ $x = \frac{3}{4}$ (1)
2. Integer (2)
   $x = 1, 3 \ldots\ldots$
3. Irrational (2)
   $x = 2$
4. Non-real (2)
   $x = 0$

c) The value of $\sqrt{33}$ lies between 2 integers. Find these integers without a calculator. (2)

$\sqrt{25}$ $\sqrt{36}$

5 & 6

[14]

QUESTION 2

The following inequality is given :

$-9 \leq 2x + 3 < 5$, $x \epsilon R$

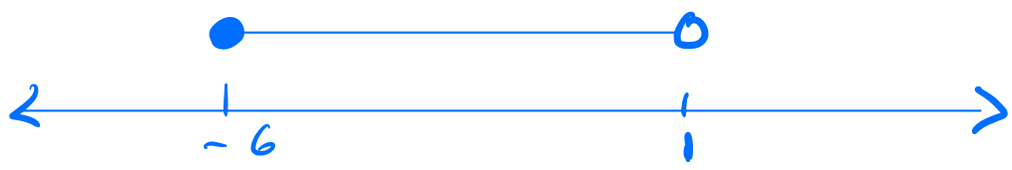
a) Solve for x. (2)

$-9 \overset{-3}{\leq} 2x+3 \overset{-3}{<} 5$

$-12 \leq 2x < 2$

$-6 \leq x < 1$

b) Represent your answer on the number line. (2)



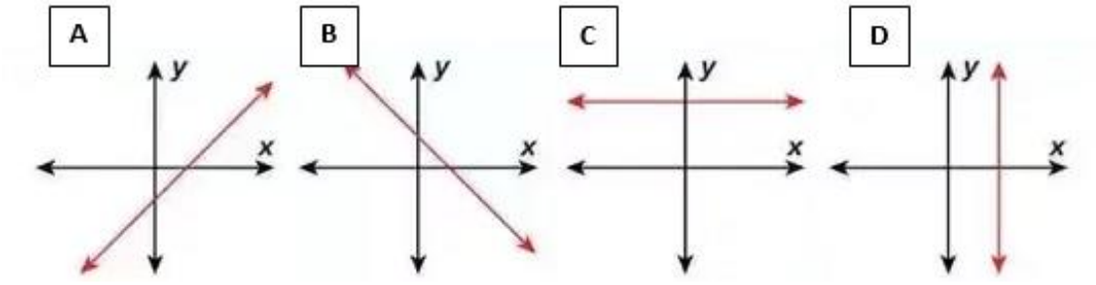
c) Write your answer in Interval Notation. (2)

$x \in [-6 ; 1)$

[6]

QUESTION 3

Write down, below each graph, if the gradient is positive , negative, equal to zero or undefined.

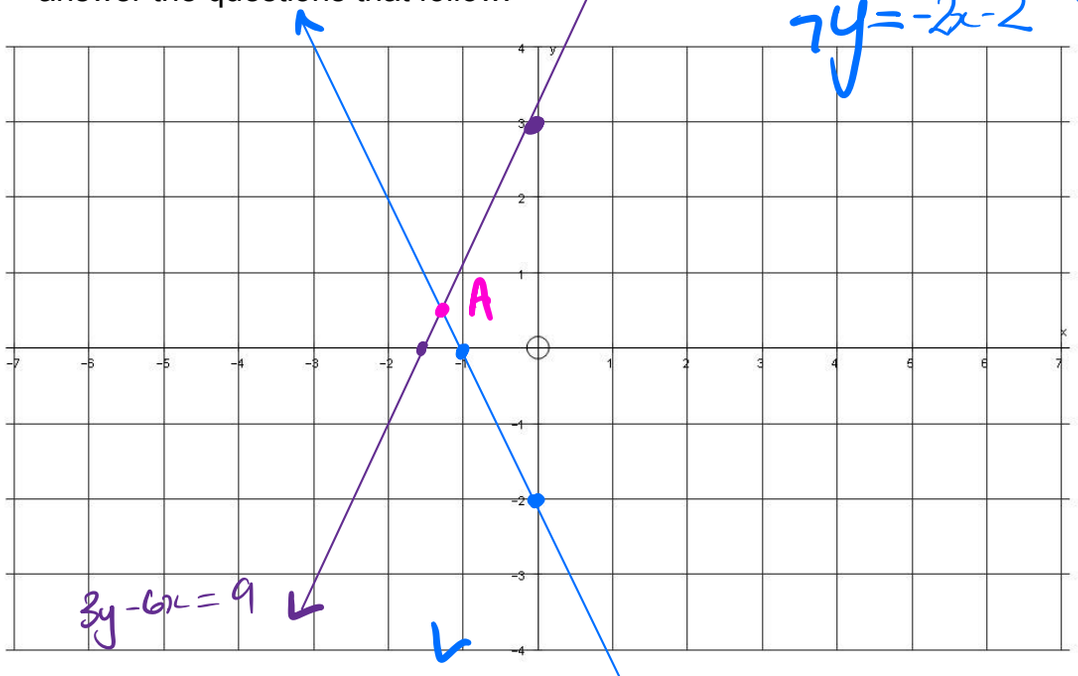


pos	neg	0	undefined

[4]

QUESTION 4

a) Sketch the graphs of $y = -2x - 2$ and $3y - 6x = 9$ on the set of axes below and answer the questions that follow.



$y = -2x - 2$

$-2x - 2 = 0$

$-2x = 2$

$x = -1$

$y = -2$

$3y = 9$

$y = 3$

$-6x = 9$

$x = \frac{9}{-6} =$

$x = -1,5$

Label your graphs and the intercepts. (6)

b) What is the equation of the y-axis?

$x = 0$ (1)

c) Label the point of intersection between the above graphs with the letter A, and write down the coordinates of the point of intersection. (2)

$(-1,2 ; 0,5)$

d) Determine the equation of a line that is parallel to $y = -2x - 2$ and has a y-intercept of 1. (2)

$m = -2$ $\quad y = -2x + 1$

e) Determine the equation of a line that passes through the points (2;5) and (-2;1) (3)

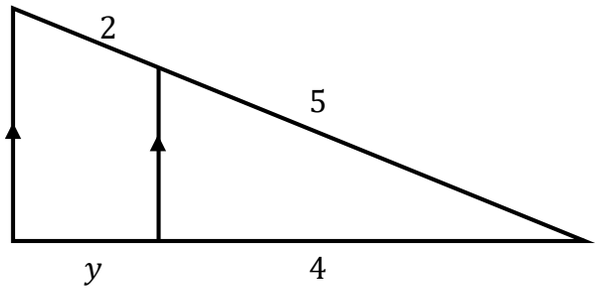
$m = \frac{1-5}{-2-2} = \frac{-4}{-4} = 1$ $\quad (2;5)$

$y = mx + c$
$5 = 1(2) + c$ $\quad c = 3$ $\quad y = 1x + 3$

[14]

QUESTION 5

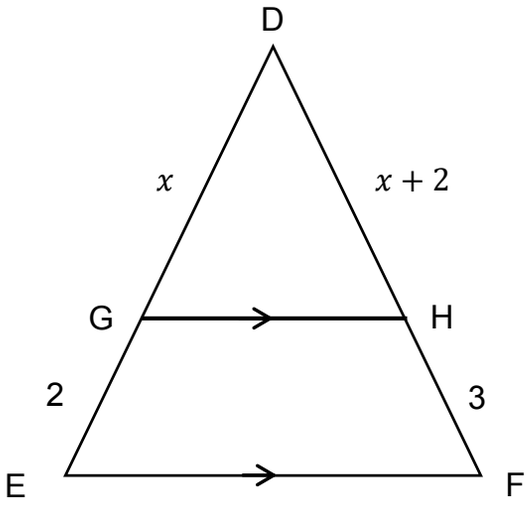
a) Find the value of y, giving reasons. (2)



$\frac{y}{4} = \frac{2}{5}$ $\quad$ (Prop Int, // lines)

$5y = 8$
$y = 1,6 \;/\; \frac{8}{5}$

b) Find the value of x, giving reasons. (3)

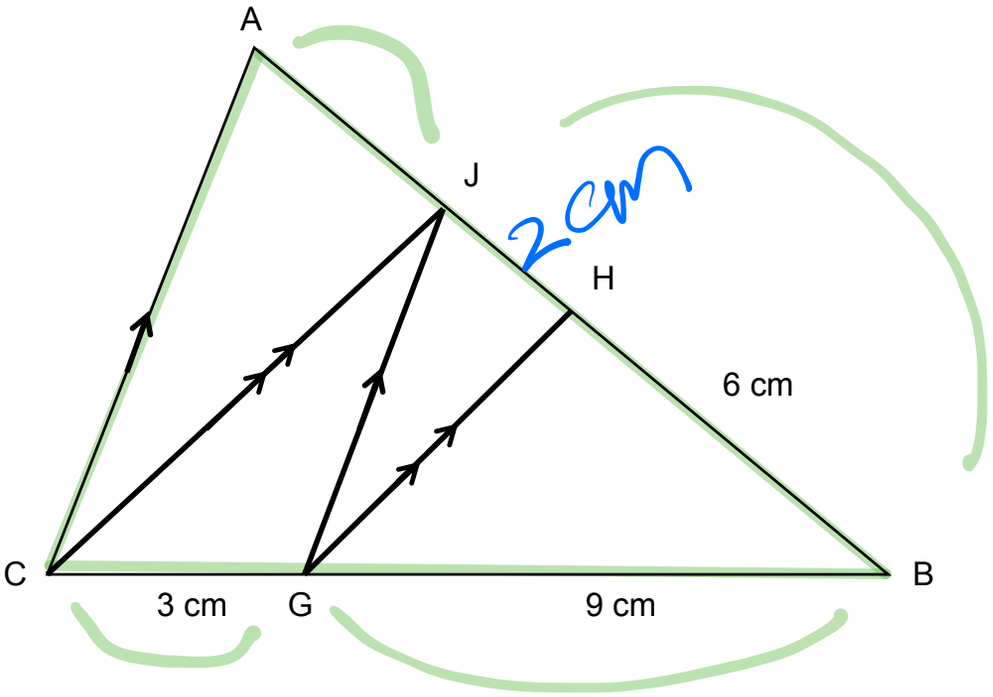


$\frac{x}{2} = \frac{x+2}{3}$ (Prop Int GH || EF)

$3x = 2x + 4$

$x = 4$

c) Given triangle ABC with AC // JG, JC // HG and GB is 9 cm, CG = 3cm and BH = 6cm.



1. Find JH, giving reasons. (3)

$\frac{JH}{6} = \frac{3}{9}$ (Prop Int Split GH)

$JH = 2$

2. Find AJ, giving reasons. (3)

$\frac{AJ}{8} = \frac{3}{9}$

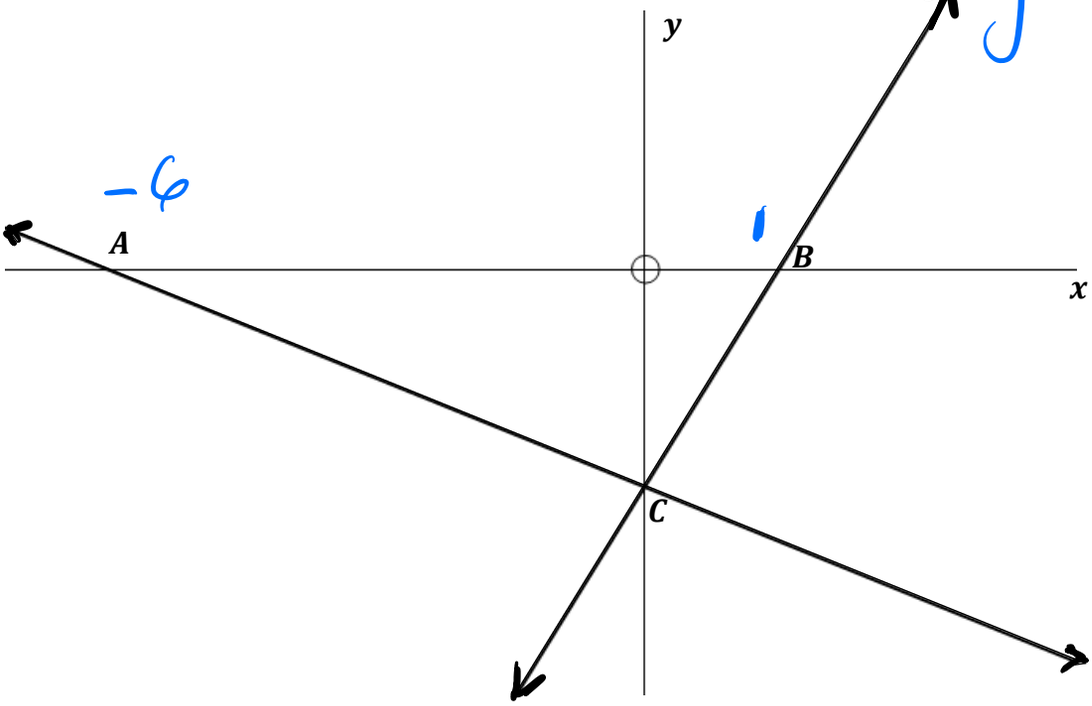
$9AJ = 24$

$AJ = \frac{8}{3}$

[11]

QUESTION 6

There are 2 graphs $y = = 2x - 3$ and $y = -\frac{1}{2}x - 3$.



a) Label the graphs on the Cartesian plane correctly (2)

b) Determine $A\hat{C}B$, giving a reason. (2)

c) Determine the length of AB (3)

d) Find the area of ΔABC. (3)

[10]

QUESTION 7

Show all working details

a) An irrational number that lies between 5 and 6 is : (2)

1) $\sqrt[3]{28}$ 3,03

2) $\sqrt{20+8}$ $\sqrt{28}=5,3$

3) $\sqrt{20\times 8}$ $\sqrt{160}=4\sqrt{10}=12,6$

4) 2π 6,3...

b) The formula for the volume of the cone is $V=\frac{1}{3}\pi r^2 h$. The radius, r, of the cone may be expressed as : (2)

1) $\sqrt{\frac{3V}{\pi h}}$

2) $\sqrt{\frac{V}{3\pi h}}$

3) $\sqrt[3]{\frac{V}{\pi h}}$

4) $\frac{1}{3}\sqrt{\frac{V}{\pi h}}$

$r^2=\frac{3V}{\pi h}$

c) If k is a negative single digit integer, which of the following is **NOT** a possible value of $\mathbf{2-k}$? (3)

1) 4

2) 8

3) 10

4) 12

$2-k=12$

$-k=10$

$k\neq -10$

[7]